

6. Project Testing Phase

6.1 Purpose of this phase

Testing confirms that the solution behaves as required, handles unexpected situations correctly and is ready for users. Test cases are derived directly from the requirements, so each requirement is proven rather than assumed.

6.2 Testing objectives

- Verify that all functional requirements are met.
- Confirm approval, rejection and exception paths.
- Validate notifications, task creation and record updates.
- Identify defects before users encounter them.
- Confirm usability and performance are acceptable.

6.3 Types of testing performed

Type	Description
Unit testing	Individual flow actions and subflows tested using Flow Designer's Test option
Functional testing	End-to-end submission, approval, task and closure
Negative testing	Invalid or missing data, rejected approvals
Integration testing	Interaction between catalog, flow, tasks and notifications
Security/role testing	Users see and do only what their roles allow
Usability testing	Ease of completing the form and tracking the request
User acceptance testing (UAT)	Business users validate against acceptance criteria

6.4 Test environment and data

- Development instance with the update set applied.
- Users: Requester (with manager), Requester (without manager), Manager, Fallback approver, Procurement user, IT Fulfilment user.
- Email logging enabled.
- Fresh requests created for each scenario so results do not overlap.

6.5 Test cases

ID	Requirement	Scenario	Steps	Expected result
TC-01	FR-01, FR-02	Catalog item is available	Log in as requester, open the portal, search for laptop	"Standard Laptop Request" appears with all defined fields
TC-02	FR-03	Mandatory field validation	Submit with an empty justification	Form blocks submission and shows a message
TC-03	FR-04, FR-05, FR-06	Valid submission	Complete all fields and submit	REQ and RITM created; flow starts; manager receives approval request
TC-04	FR-07, FR-09	Manager approves	Approve as manager	Procurement task created for Procurement Team; requester notified
TC-05	FR-08, FR-09	Manager rejects	Reject with comment	RITM closed incomplete; reason recorded; requester notified
TC-06	FR-10, FR-11	Procurement completion	Close the procurement task	IT setup task created; RITM state updated
TC-07	FR-10, FR-11	Fulfilment completion	Close the IT setup task	RITM closed complete; completion notification sent
TC-08	Business rule 3	No manager on user	Submit for a user with no manager	Request routed to fallback approver
TC-09	FR-12	Approval not answered	Leave approval pending beyond the reminder period	Reminder email sent to approver
TC-10	Non-functional	Past needed-by date	Enter yesterday's date	Client script prevents submission
TC-11	Security	Unauthorized approval	Attempt approval as unrelated user	Action not permitted
TC-12	Error handling	Missing procurement group	Temporarily unset group and submit	Flow error captured; admin notified; RITM flagged
TC-13	FR-13	Reporting	Run dashboard after several requests	Counts and durations reflect test data

6.6 Test execution process

1. Prepare the scenario and data.
2. Execute the steps exactly as written.
3. Observe results in the portal, the RITM, the approval record, the task and the **Flow Designer execution log**.
4. Compare the actual result with the expected result.
5. Record status: *Pass*, *Fail* or *Blocked*, with screenshots.
6. Log any failure as a defect with steps to reproduce.

6.7 Test result recording template

Test ID	Date	Tester	Actual result	Status	Defect ID
TC-01					
TC-02					
...					

6.8 Defect management

Field	Description
Severity	Critical, High, Medium, Low
Status	New, In progress, Fixed, Retested, Closed
Details	Steps to reproduce, expected vs actual, evidence

Typical issues found in solutions like this, and their usual fixes, include: approval not triggering because the approver field was blank (fix: fallback logic), notification not showing dynamic values (fix: correct data pill mapping), and wait conditions never resolving because the wrong task state was checked (fix: correct condition and retest).

6.9 Regression testing

After each fix, the happy path (TC-03 to TC-07) is executed again to confirm that no earlier behaviour was broken.

6.10 Performance and reliability checks

- Confirm the flow starts promptly after submission.
- Submit several requests in succession and verify that all execute correctly.
- Review the execution log for errors or long-running items.

6.11 User acceptance testing

Business users (a requester, a manager and members of Procurement and IT) run through realistic scenarios and confirm:

- The form is easy to complete.
- The approval request is clear.
- The tasks contain enough detail to act on.
- Notifications are understandable and timely.

Feedback is recorded and any agreed changes are applied before sign-off.

6.12 Exit criteria

Testing is complete when:

- All high-priority test cases have passed.
- No critical or high severity defects remain open.
- Regression testing is successful.
- UAT sign-off is obtained.

6.13 Outcome of the phase

A verified solution supported by documented test cases, execution evidence, resolved defects and user acceptance. The team can now finalize documentation and prepare for demonstration.